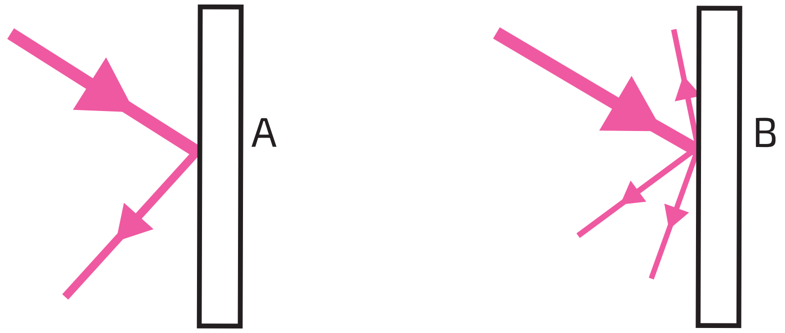
Reflection Practice

1 The diagrams below show the two ways light can reflect.



(a) Which diagram shows specular reflection?

(b) Which diagram shows diffuse reflection?

2 Which kind of reflection is described in each case? Fill in the table

Description	Specular or Diffuse?
Seeing your face in a polished metal spoon	
Light bouncing off a sheet of printer paper	
Light bouncing off a dry concrete pavement	
Sunlight reflecting off a perfectly still puddle of water	

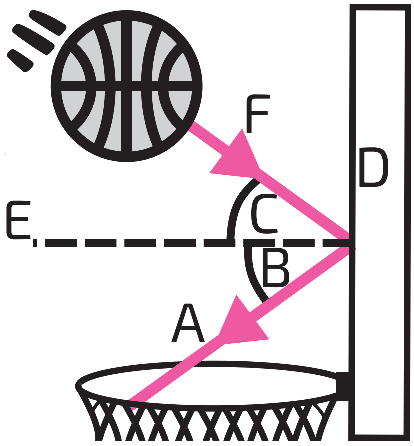
3 Your housemate always takes ages in the bathroom and you suspect they're spending a long time staring at themselves in the mirror. You 'clean' the mirror with a dirty cloth and now reflections are blurred. What kind of reflection is this?

4 The brightness of reflected rays depend on the type of reflection. Complete the sentences using the words **rough**, **diffuse**, **scattered**, **smooth**, **specular** and **direction**.

In a large lecture theatre, the back walls are often fitted with ________, jagged acoustic panels to encourage ________ reflection. This ensures that sound is ________ randomly around the room, preventing a distracting, bouncing echo.

On the other hand, in a mosque, the ________, curved section of the wall called the mihrab is designed for ________ reflection. This acts like a mirror for sound, focusing the speaker's voice in a specific ________ toward the worshippers so that the message remains loud and clear.

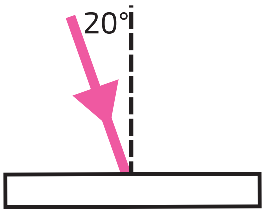
5 The diagram below shows a basketball being shot off a backboard. Which words describe the features of the diagram? Fill in the table.



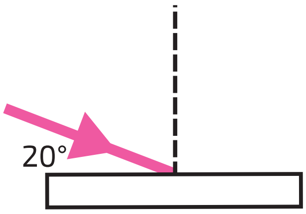
Technical term	A, B, C, D, E or F?
surface	
reflected ray	
incident ray	
angle of reflection	
angle of incidence	
normal	

6 In each diagram below, write down the angle of incidence.

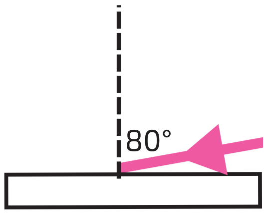
(a)



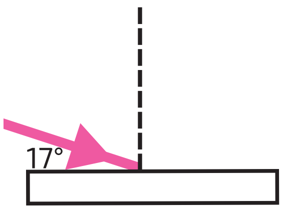
(b)



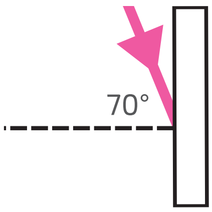
(c)



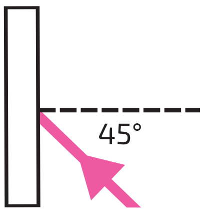
(d)



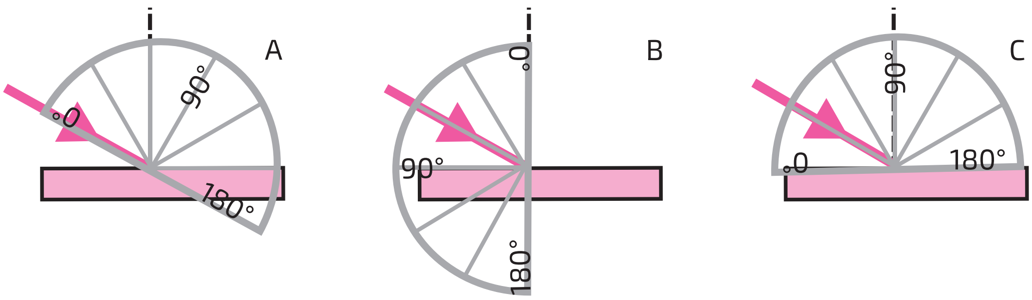
(e)



(f)



7 The diagrams below show three attempts to measure an angle of incidence for the same light ray.

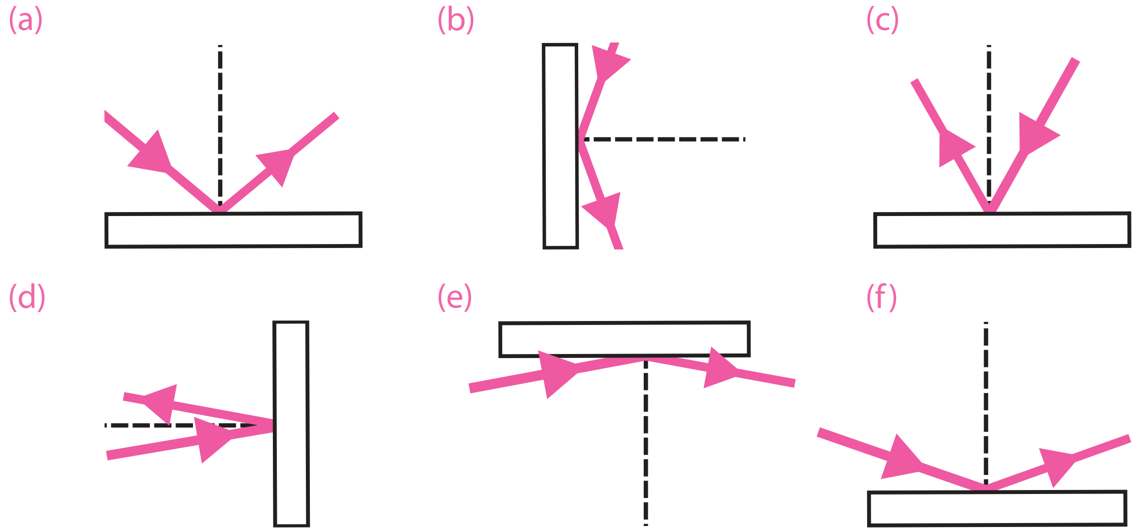


(a) From which of these protractors could we **not** read off the correct angle?

(b) State the angle of incidence of the ray.

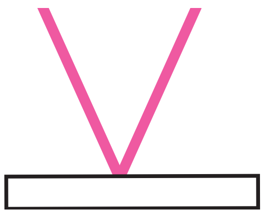
(c) What do we line the centre of the protractor up with?

8 Measure the angles of incidence and the angles of reflection using a protractor.



9 An experiment is set up where a laser is reflected off a surface then absorbed by a receiver. Due to the camera angle, you cannot see either the laser emitter or receiver. The laser is turned on the entire time.

- **Video 1**: The original recording.
- **Video 2**: The original recording, but played in reverse.
- **Video 3**: A new recording where the laser pointer and target have swapped sides.



The camera is zoomed so that you can only see the beam and the mirror — not the pointer or the target. Is it possible to see the difference between the videos? Why?

10 A ray is fired at two surfaces with the same angle of incidence each time. Surface A reflects all the energy, while Surface B absorbs $30\ \%$ of it and reflects the rest. The reflection is specular in both cases. What can we say about the angle of reflection for each surface?

11 Four large vertical mirrors are set up in a square shape. A small laser pointer is attached to one of the mirrors and pointed at another with an angle of incidence of $45°$. What shapes could the laser trace as it bounces off the mirrors with this set up?